System Al–Fe

Solution Phases:

(stable)	:	Liquid, fcc–$A1$, bcc–$A2$, bcc–$B2$,Al_5Fe_4
(metastable)	:	cbcc–$A12$, cub–$A13$, hcp–$A3$

Compounds:

stable	:	Al_2Fe, Al_5Fe_2,$Al_{13}Fe_4$,
metastable	:	$Al_{11}Mn_4$, $Al_{12}Mn$, Al_6Mn

Modelling:

Liquid	:	Substitutional, Redlich–Kister
cbcc–$A12$	:	Substitutional, Redlich–Kister
cub-A13	:	Substitutional, Redlich–Kister
fcc–$A1$	:	Substitutional, Redlich–Kister
hcp–$A3$	:	Substitutional, Redlich–Kister
bcc–$A2$	:	see below
bcc–$B2$	:	Sublattice model, $(Al,Fe)_{0.5}(Al,Fe)_{0.5}(\square)_3$ Two descriptions of the bcc–$A2$:bcc–$B2$ order-disorder transformation are given: one which includes the parameters of the order and disordered phases in a single description (equation 21 in the thermodynamic modelling chapter), and a second one where the order and disorder contributions are expressed by equation 23. For the second case, the parameters are in a frame.
Al_5Fe_4	:	Substitutional, Redlich–Kister
$Al_{13}Fe_4$	:	Sublattice model , $(Al)_{0.6275}(Fe)_{0.235}(Al,\square)_{0.1375}$
Al_2Fe	:	Stoichiometric, $(Al)_2(Fe)$
Al_5Fe_2	:	Stoichiometric, $(Al)_5(Fe)_2$
$Al_{11}Mn_4$	:	Stoichiometric, $(Al)_{11}(Mn)_4$
$Al_{12}Mn$	:	Stoichiometric, $(Al)_{12}(Mn)$
Al_6Mn	:	Stoichiometric, $(Al)_6(Mn)$

Assessor and Date:

M. Seierstein, 1991

Comments:

$Al_{11}Mn_4$, $Al_{12}Mn$ and Al_6Mn and the hcp–$A3$ phase have been evaluated by Å. Jansson and T.G.Chart (1995) in the assessment of the Al-Fe-Mn system.

Thermodynamic properties of the solution and compound phases (J.mol^{-1})

Phase fcc–$A1$

$$L^{0,\mathrm{fcc}-A1}_{\mathrm{Al,Fe:\square}} = -76066.1 + 18.6758\ T$$

$$L^{1,\mathrm{fcc}-A1}_{\mathrm{Al,Fe:\square}} = 21167.4 + 1.3398\ T$$

Phase hcp–$A3$

$$L^{0,\mathrm{hcp}-A3}_{\mathrm{Al,Fe:\square}} = -106903.0 + 20.0\ T$$

Phase liquid

$$L^{0,\mathrm{liquid}}_{\mathrm{Al,Fe}} = -91976.5 + 22.1314\ T$$

$$L^{1,\mathrm{liquid}}_{\mathrm{Al,Fe}} = -5672.58 + 4.8728\ T$$

$$L^{2,\mathrm{liquid}}_{\mathrm{Al,Fe}} = 121.9$$

Phase Al_2Fe

$$G^{\circ}(T) - 2.0\ H^{\circ,\mathrm{fcc}-A1}_{\mathrm{Al}}(298.15\ \mathrm{K}) - H^{\circ,\mathrm{bcc}-A2,\mathrm{para}}_{\mathrm{Fe}}(298.15\ \mathrm{K}) = -98097.0 + 18.7503\ T + 2.0\ \mathrm{GHSER_{Al}} + \mathrm{GHSER_{Fe}}$$

Phase $Al_{13}Fe_4$

$$G^{\circ}(T) - 0.765\ H^{\circ,\mathrm{fcc}-A1}_{\mathrm{Al}}(298.15\ \mathrm{K}) - 0.235\ H^{\circ,\mathrm{bcc}-A2,\mathrm{para}}_{\mathrm{Fe}}(298.15\ \mathrm{K}) = \mathrm{G(Al{:}Fe{:}Al)} = -30714.4 + 7.44\ T + 0.765\ \mathrm{GHSER_{Al}} + 0.235\ \mathrm{GHSER_{Fe}}$$

$$G^{\circ}(T) - 0.6275\ H^{\circ,\mathrm{fcc}-A1}_{\mathrm{Al}}(298.15\ \mathrm{K}) - 0.235\ H^{\circ,\mathrm{bcc}-A2,\mathrm{para}}_{\mathrm{Fe}}(298.15\ \mathrm{K}) = \mathrm{G(Al{:}Fe{:}\square)} = -27781.3 + 7.2566\ T + 0.6275\ \mathrm{GHSER_{Al}} + 0.235\ \mathrm{GHSER_{Fe}}$$

Phase Al_5Fe_2

$$G^{\circ}(T) - 5.0\ H^{\circ,\mathrm{fcc}-A1}_{\mathrm{Al}}(298.15\ \mathrm{K}) - 2.0\ H^{\circ,\mathrm{bcc}-A2,\mathrm{para}}_{\mathrm{Fe}}(298.15\ \mathrm{K}) = -228576 + 48.99503\ T + 5.0\ \mathrm{GHSER_{Al}} + 2.0\ \mathrm{GHSER_{Fe}}$$

Phase Al_5Fe_4

$G^o(T) - H^{o,fcc-A1}_{Al}(298.15\ K) = 12178.90 - 4.813\ T + GHSER_{Al}$

$G^o(T) - H^{o,bcc-A2,para}_{Fe}(298.15\ K) = 5009.03 + GHSER_{Fe}$

$L^{0,Al_5Fe_4}_{Al,Fe} = -131649 + 29.4833\ T$

$L^{1,Al_5Fe_4}_{Al,Fe} = -18619.5$

Phase bcc–$B2$

$G^o(T) - H^{o,fcc-A1}_{Al}(298.15\ K) = G(Al{:}Al{:}\square) = GBCC_{Al}$

$G^o(T) - 0.5\ H^{o,fcc-A1}_{Al}(298.15\ K) - 0.5\ H^{o,bcc-A2,para}_{Fe}(298.15\ K) = G(Fe{:}All{:}\square) =$
$-37890.5 + 7.9972\ T + 0.5\ GBCC_{Al} + 0.5\ GHSER_{Fe}$

$T^{bcc-B2}_{c,Fe:Al:\square} = 521.5$ $\qquad \beta^{bcc-B2}_{Fe:Al::\square} = 1.11$

$G^o(T) - 0.5\ H^{o,fcc-A1}_{Al}(298.15\ K) - 0.5\ H^{o,bcc-A2,para}_{Fe}(298.15\ K) = G(Al{:}Fe{:}\square) =$
$-37890.5 + 7.9972\ T + 0.5\ GBCC_{Al} + 0.5\ GHSER_{Fe}$

$T^{bcc-B2}_{c,Al:Fe:\square} = 521.5$ $\qquad \beta^{bcc-B2}_{c,Al:Fe:\square} = 1.11$

$G^o(T) - H^{o,bcc-A2,para}_{Fe}(298.15\ K) = G(Fe{:}Fe{:}\square) = GHSER_{Fe}$

$T^{bcc-B2}_{c,Fe:Fe:\square} = 1043K$ $\qquad \beta^{bcc-B2}_{c,Al:Fe:\square} = 2.22$

$L^{0,bcc-B2}_{Al,Fe:Al:\square} = L^{0,bcc-B2}_{Al:Al,Fe:\square} = -22485.07 + 7.9772\ T$

$L^{1,bcc-B2}_{Al,Fe:Al:\square} = L^{1,bcc-B2}_{Al:Al,Fe:\square} = 368.15$

$T^{0,bcc-B2}_{c,Al,Fe:Al:\square} = T^{0,bcc-B2}_{c,Al:Al,Fe:\square} = 189.0$

$T^{1,bcc-B2}_{c,Al,Fe:Al:\square} = T^{1,bcc-B2}_{c,Al:Al,Fe:\square} = 63.0$

$\beta^{0,bcc-B2}_{Al,Fe:Al:\square} = \beta^{0,bcc-B2}_{Al:Al,Fe:\square} = 0.0$

$T^{0,bcc-B2}_{c,Al,Fe:Fe:\square} = T^{0,bcc-B2}_{c,Fe:Al,Fe:\square} = -189.0$

$T^{1,bcc-B2}_{c,Al,Fe:Fe:\square} = T^{1,bcc-B2}_{c,Fe:Al,Fe:\square} = 63.0$

$\beta^{0,bcc-B2}_{Al,Fe:Fe:\square} = \beta^{0,bcc-B2}_{Fe:Al,Fe:\square} = 0.0$

$L^{0,bcc-B2}_{Fe:Al,Fe:\square} = L^{0,bcc-B2}_{Al,Fe:Fe:\square} = -24694.0 + 7.9772\ T$

$L^{1,bcc-B2}_{Fe:Al,Fe:\square} = L^{1,bcc-B2}_{Al,Fe:Fe:\square} = 368.15$

$G^{*}_{\mathrm{Al:Fe:\square}} = -\ 860\ R \qquad (R = 8.31451)$

$L^{0}_{\mathrm{Al,Fe}} = -\ 30740 + 7.9972\ T$

$L^{1}_{\mathrm{Al,Fe}} = 386.15$

Phase bcc–*B*2

$G^{\circ}(T) - H^{\circ,\mathrm{fcc}-A1}_{\mathrm{Al}}(298.15\ \mathrm{K}) = \mathrm{G(Al:Al:\square)} = 0.0$

$G^{\circ}(T) - 0.5\ H^{\circ,\mathrm{fcc}-A1}_{\mathrm{Al}}(298.15\ \mathrm{K}) - 0.5\ H^{\circ,\mathrm{fcc}-A1,\mathrm{para}}_{\mathrm{Fe}}(298.15\ \mathrm{K}) = \mathrm{G(Al:Fe:\square)} = 2.0\ G^{*}_{\mathrm{Al:Fe:\square}}$

$G^{\circ}(T) - 0.5\ H^{\circ,\mathrm{fcc}-A1}_{\mathrm{Al}}(298.15\ \mathrm{K}) - 0.5\ H^{\circ,\mathrm{fcc}-A1,\mathrm{para}}_{\mathrm{Fe}}(298.15\ \mathrm{K}) = \mathrm{G(Fe:Al:\square)} = 2.0\ G^{*}_{\mathrm{Al:Fe:\square}}$

$G^{\circ}(T) - H^{\circ,\mathrm{fcc}-A1,\mathrm{para}}_{\mathrm{Fe}}(298.15\ \mathrm{K}) = \mathrm{G(Fe:Fe:\square)} = 0.0$

Phase bcc–*A*2

$L^{0,\mathrm{bcc}-A2}_{\mathrm{Al,Fe:\square}} = 4.0\ \ L^{0}_{\mathrm{Al,Fe}}$

$L^{1,\mathrm{bcc}-A2}_{\mathrm{Al,Fe:\square}} = 8.0\ \ L^{1}_{\mathrm{Al,Fe}}$

$T^{1,\mathrm{bcc}-A2}_{\mathrm{C,Al,Fe:\square}} = 504$

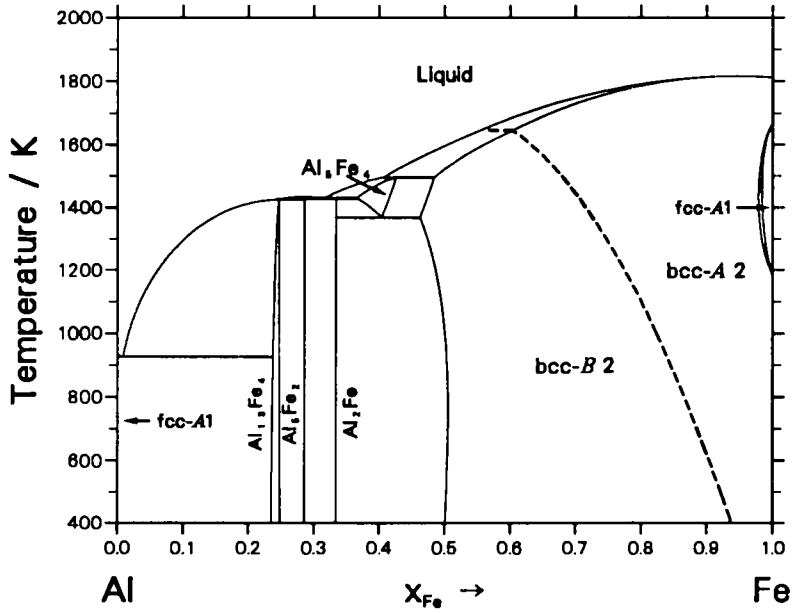

Table I - Invariant Reactions.

Reaction	Type	Compositions x_{Fe}			T / K
$Al_5Fe_4 \rightleftharpoons$ bcc-$B2$ + Al_2Fe	Eutectoid	.403	.462	.333	1368.2
$Al_5Fe_2 \rightleftharpoons$ Liquid	Congruent	.286	.286	....	1434.5
Liquid $\rightleftharpoons Al_5Fe_2 + Al_5Fe_4$	Eutectic	.316	.286	.365	1430.0
Liquid + $Al_5Fe_2 \rightleftharpoons Al_{13}Fe_4$	Peritectic	.242	.286	.247	1424.4
Liquid $\rightleftharpoons$ fcc-$A1$ + $Al_{13}Fe_4$	Eutectic	.009	.000	.237	927.1
Liquid + bcc-$B2 \rightleftharpoons Al_5Fe_4$	Peritectic	.405	.483	.425	1495.5
$Al_5Fe_2 + Al_5Fe_4 \rightleftharpoons Al_2Fe$	Peritectoid	.286	.366	.333	1428.1

Table II - Crystal Structure and Phase Description.

System	Phase	Prototype	Pearson Symbol Space Group	Sub-lattices	Comments
Al-Fe	(Al)	Cu	$cF4$ $Fm\bar{3}m$	M 4	
	(Fe)	Cu	$cF4$ $Fm\bar{3}m$	M 4	
	(Fe)	W	$cI2$ $Im\bar{3}m$	M 2	
	AlFe	CsCl	$cP2$ $Pm\bar{3}m$	M_1 1 M_2 1	
	$AlFe_3$	BiF_3	$cF16$ $Fm\bar{3}m$	Al 4 Fe_1 4 Fe_2 8	not considered
	Al_2Fe	Al_2Fe	$aP18$ $P1$		
	Al_3Fe		$mC102$ $C2/m$		designated by $Al_{13}Fe_4$
	Al_5Fe_2		oC^* $Cmcm$		
	ϵ		$cI16$?		designated by Al_5Fe_4
	Al_6Fe	Al_6Fe	$oC28$ $Cmc2_1$	Al_1 4 Al_2 4 Al_3 8 Al_4 8 Fe 4	metastable
	Al_9Fe_2	Al_9Co_2	$mP22$ $P2_1/c$		metastable